Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience building and shipping **LLM agent** and **RAG** systems end to end (React, FastAPI, AWS) with **Azure OpenAI**, **LlamaIndex** and **pgvector**, first as the sole engineer alongside my manager, later as de facto team lead for up to 4 interns. Owns the whole loop: chooses models on cost and fit (self-hosted **Llama-3** on a GPU instance for the largest department, Azure OpenAI for decision makers), writes and versions the system prompts, designs the tool-calling agents and the retrieval context behind them, ships to real users and changes the system based on what they find. Placed 2nd at the Encode Vibe Coding Hackathon building a working app from a spec with codeplain's spec-driven framework, and uses **Claude Code**. Completed an MSc in Data Science (awaiting result) with a dissertation project with AstraZeneca; twice recognised with Best Team Project.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Agent, Context & Model Layer in Production

- Specified, designed and built the first GenAI systems at the company end to end, initially with only my manager: **RAG** applications (ReactJS frontend, **Python FastAPI** REST API microservices, **LlamaIndex** with **Azure OpenAI** GPT-3.5 & Llama-3) for Compliance, Client Service and Investment. Built the context layer with **Azure OpenAI embeddings** and a separate **pgvector** database per department, so each team retrieved only from its own internal documents.
- Built **tool-calling** agents that call **Python** functions as tools to run **SQL**-backed fund comparisons and generate PDF reports on request; steered behaviour with **system prompts** versioned in **Git**, released to real users, changed the applications based on what they found, and had the Compliance department review each release.
- Made the **model selection** call on cost against quality: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and the stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models as needs changed.

Ownership & Delivery

- Worked directly with Client Service, Compliance and Investment to turn requests into clear requirements, demonstrate working tools and turn feedback into releases; acted as de facto team lead for up to 4 interns in my final year; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).
- Built a **Python** pipeline, scheduled with **EventBridge**, **Lambda** and **AWS SageMaker**, that uses **Azure OpenAI** to parse over **300** daily emails from credit rating agencies into structured data (first version used **AWS Textract**, replaced when a rating agency began sending emails as images), fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60** analyst hours a month.
- Migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10** hours a week, then recreated them in the **ReactJS** frontend because Power BI licensing was too costly to scale to the nationwide Client Service team.

Secure Production Delivery & Data Foundations

- Turned prototypes into deployable tools: packaged the **FastAPI** microservices (uvicorn) as **Docker** containers on **AWS** (EC2, ECR) and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Protected the internal applications with **RBAC** using a **Python** authentication function against the company **LDAP** server and **AWS Security Groups** limiting access to company private IPs; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy analyst data.

PROJECTS

- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes, stored as data so tuning a persona needs no code change, and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. Data preprocessing pipeline, scoring algorithm and the RAG-based AI assistant that narrates results and answers methodology questions, in a tool built as FastAPI microservices with a React frontend that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/

ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Claude Code (AI-assisted development), codeplain (spec-driven development), Azure OpenAI, LlamaIndex, REST APIs, Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Nginx, Linux, LDAP, Power BI, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite